Welcome to the Course

PNAP0143 Logical Reasoning

(Penaakulan Mantik)

MHA

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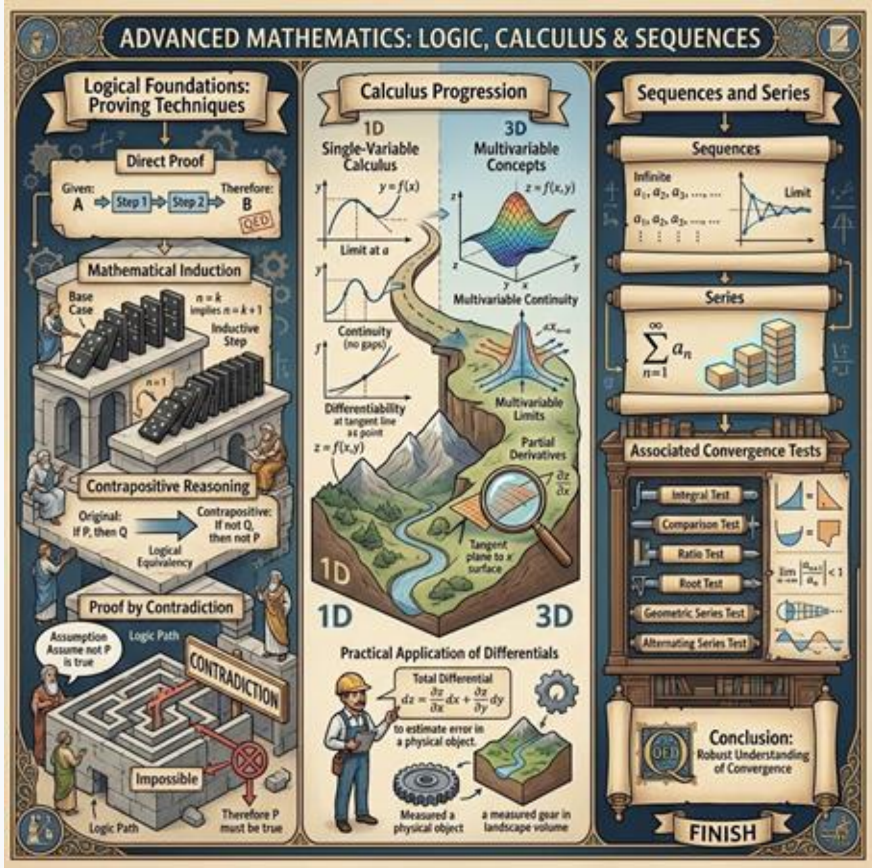
Course Synopsis



This course employs an integrated methodology that combines calculus with rigorous mathematical proof, prioritizing conceptual depth and analytical problem-solving skills.

The curriculum is structured around the following key areas:

- **Logical Foundations: A deep dive into essential proving techniques, including direct proof, mathematical induction, contrapositive reasoning and proof by contradiction.**
- **Calculus Progression: An advancement from single-variable calculus to multivariable concepts, establishing a robust understanding of limits, continuity and differentiability, alongside the practical application of differential.**
- **Sequences and Series: A comprehensive examination of sequences, series and their associated convergence tests to conclude the course.**

Curriculum Outcomes

Course Learning Outcomes

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Mastering mathematical logic and calculus applications across four core pillars.

CLO 1 • Mathematical Logic

Proving Techniques

Identify and master various rigorous techniques of proving in mathematical logic, laying the foundation for analytical reasoning.

CLO 2 • Real Functions

Single & Multivariable

Apply key concepts of real functions across both single and multivariable scenarios to model real-world continuous phenomena.

CLO 3 • Analysis Foundations

Limits & Derivatives

Relate and apply the core mathematical principles of limits, continuity, and derivatives to solve dynamic rate-of-change problems.

CLO 4 • Advanced Calculus

Sequences & Series

Explain the fundamental properties of infinite sequences and series, and analyze their behaviors and convergence criteria.

17 Weeks Syllabus Progression

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Week 1	**Logic & Truth Tables:** Week 1 introduces logic, including truth tables for connectives and quantifiers.
Week 2	**Methods of Proof:** Week 2 covers methods of proof, such as direct proof, contarpositive, proof by contradiction and mathematical induction.
Week 3	**Real Numbers & Inequalities:** Week 3 explores real numbers and inequalities.
Weeks 4–5	**Functions & Graphs:** Weeks 4 and 5 focus on functions and graphs, including symmetry and function transformations.
Weeks 6–7	**Limits & Asymptotes:** Weeks 6 and 7 dive into calculating limits, evaluating one-sided limits and identifying vertical and horizontal asymptotes.

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17 Weeks Syllabus Progression

Week 8	**Continuity & Squeeze Theorem:** Structured study of limits, continuity definitions, and the mathematical Squeeze Theorem.
Weeks 9-12	**Differentiation & Optimization:** Comprehensive coverage of core differentiation rules, L'Hopital's rule, and practical applications in optimization and concavity.
Week 13	**Related Rates:** Practical calculus applications analyzing dynamic related rates of change.
Week 14	**Conic Sections:** Geometric definitions, equations, and applications of classic conic sections.
Week 15	**Two Variables:** Introductory study of multi-variable mathematics and functions of two variables.
Weeks 16-17	**Sequences & Series:** Final academic block exploring sequence theory, series convergence tests, and Maclaurin series expansions.

Course Grading Structure

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Continuous assessments and a comprehensive final exam account for half of your total grade each

Continuous

Online Quizzes

10%

This includes Quiz 1 and Quiz 2, each worth 5% of your total grade to evaluate early course concepts.

Continuous

Group Assignments

30%

There are two major group assignments, Assignment 1 and Assignment 2, each contributing 15% to your final score.

Continuous

Mid-term Test

10%

The physical Mid-term Test provides another 10% towards your grade, verifying core semester materials in person.

Summative

Final Examination

50%

The remaining 50% of your grade is determined by the Final Examination, which evaluates all comprehensive topics covered throughout the semester.

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Assessments: Quizzes & Mid-term Test

Assessment Platform: UKMFolio

Online Quizzes

Students will complete two online quizzes via UKMFolio. Each quiz consists of 10 multiple-choice questions worth 10 marks.

Quiz 1: 7th August*
10 multiple-choice questions • 10 marks

Quiz 2: 2nd November*
10 multiple-choice questions • 10 marks

Assessment Venue: Physical

Mid-term Test

The Mid-term Test is a physical examination held between August 24 and August 28. It comprises 4 to 5 structured questions totaling 40 marks.*

Schedule: 24th August – 28th August
Physical exam format

Scope: Week 1 through Week 7
Providing a comprehensive review of all topics covered

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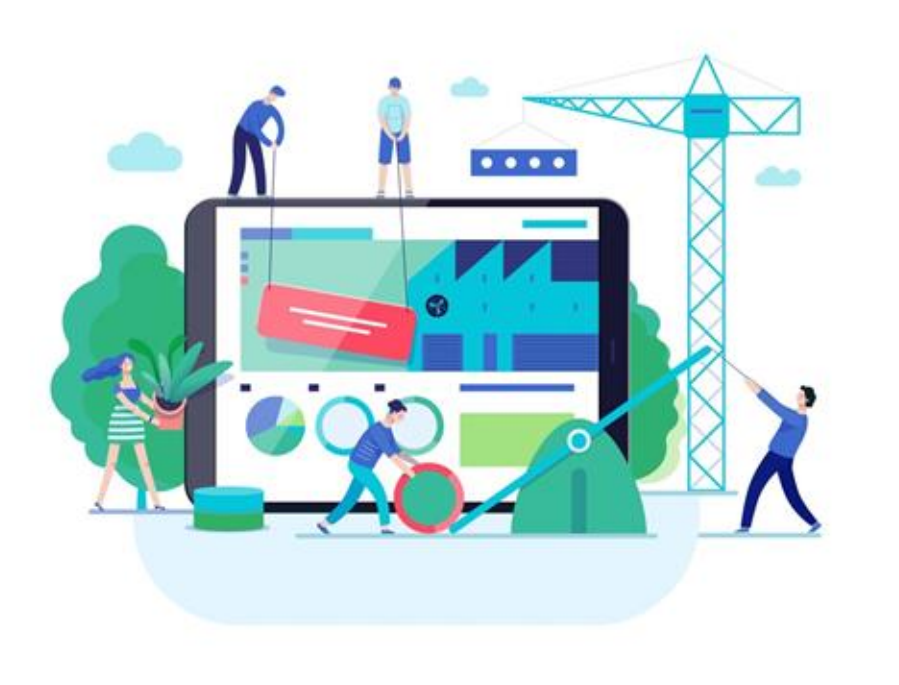
Collaborative Coursework (Group Assignment)

Collaborative work is a key component of this course, evaluated through two major assignments designed to promote peer learning and problem-solving.



Assignment 1 Groups of 2 to 3 Students

Requires groups of 2 to 3 students to solve 10 structured questions targeting core syllabus concepts.*

Assignment 2 Groups of 4 to 5 Students

Involves larger groups of 4 to 5 students tackling 10 structured questions covering differentiation, rate of change, and continuity.*

Artificial Intelligence Integration

Students are expected to responsibly utilize AI tools to assist in solving complex multivariable and optimization problems.

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Final Examination & Revision Week

Course Assessment

Final Examination

Weight & Marks: 50% of total grade | 80 marks*

Format: 5 structured questions*

Location: Physical assessment held at DTAMS

Date: 16th Nov – 20th Nov

Scope: Comprehensive review of Weeks 1 to 14 course topics

Preparation Window

Dedicated Revision Week: November 9 – November 13

Students are highly encouraged to utilize this period for independent preparation and independent learning review to ensure comprehensive success before the final examination period.

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